

PIXNET: Powering social and image analysis with Google Cloud Platform

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ABOUT PIXNET

Using Google Cloud Platform, PIXNET delivers cloud-based personalization for social and targeting capabilities for advertisers. The business is also built on its agility, scalability, response times, and positioning for growth.

PIXNET

About PIXNET

Founded in 2003 and based in Taiwan, PIXNET allows people to share photos, create blogs, and communicate online or via mobile applications. The website is one of the most visited sites in the country.

Industries: Technology

Location: Taiwan

Cloud SQL (<https://cloud.google.com/sql/>),

Google Cloud results

- Reduces response times for customers from 50 milliseconds to 10 milliseconds
- Enables deeper data analysis to deliver personalization and targeting
- Accelerates deployment cycles and improves reliability and agility

Social networks are now well established as enabling people to connect and share. In Taiwan,

PIXNET

(<https://www.pixnet.net/>)

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Enabling a class of 18 image cust busi serv

Google Cloud (<https://cloud.google.com/>)
(<https://cloud.google.com/vpc/>), [Cloud](https://cloud.google.com/composer/)

Kubernetes Engine
(<https://cloud.google.com/kubernetes-engine/>)
(<https://cloud.google.com/composer/>)

Compute Engine
(<https://cloud.google.com/compute/>)

Cloud Storage (<https://cloud.google.com/storage/>)
(<https://workspace.google.com/products/gmail/>)

Cloud SQL (<https://cloud.google.com/sql/>)

Virtual Private Cloud
(<https://cloud.google.com/vpc/>)

Cloud Pub/Sub (<https://cloud.google.com/pubsub/>)
(<https://workspace.google.com/products/drive/>)

Cloud Composer
(<https://cloud.google.com/composer/>)

BigQuery (<https://cloud.google.com/bigquery/>)

Vision AI (<https://cloud.google.com/vision/>) [lp.](https://workspace.google.com/products/gmail/)

Google Workspace
(<https://workspace.google.com/>)

Gmail
(<https://workspace.google.com/products/gmail/>)

Calendar
(<https://workspace.google.com/products/calendar/>)

Google Drive
(<https://workspace.google.com/products/drive/>)

communicate
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mobile
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Tseng, Manager
of Platform
Software
Engineering at
PIXNET.

The business aggregates Chinese-language lifestyle content onto its platform. Customers can join the service to access the content and create groups that share similar likes and interests. “Our lifestyle content platform has become a ‘must-visit’ website for people considering buying goods and services,” says Tseng.

Once people have reviewed comments and recommendations on the platform, they can proceed to purchase by clicking links on the PIXNET ecommerce platform, PIXgoods.

PIXNET also analyzes data collected when users interact, locate, browse, or purchase on its platforms. The resulting insights inform marketing tools that help businesses target users with relevant advertising.

AI and ML power precise results

The emergence of artificial intelligence and machine learning is paving the way for PIXNET to conduct deeper analysis and obtain more precise results.

The business can then use these outcomes to improve its services. “Our vision—that we are executing now—is to use computation algorithms to enable advertisers to target users with relevant products and services and to provide a personalized experience to users,” explains Tseng.

The rapid growth of the Internet of Things—following the proliferation of smartphones—also presents PIXNET with opportunities and challenges in delivering its products and services to customers.

“The emergence of interactive speakers and similar devices is prompting us to consider new ways of approaching customers globally,” says Tseng.

Since it was founded in 2003, PIXNET had run its platforms and applications from physical infrastructure in a data center. However, the infrastructure did not support its need to be agile and responsive and to dedicate valuable internal resources and personnel to product development rather than administration and maintenance.

In this traditional environment, building production environments took up to three days, while constraints on bandwidth and hard disk capacity could cause service interruptions during periods of peak demand.

Its existing data center arrangements meant PIXNET had to predict its likely future bandwidth needs and did so based on its previous years' usage. However, extraordinary demand meant the business soon exceeded the purchased bandwidth.

Moving to Google Cloud

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—Arrack Tseng, Manager,
Platform Software
Engineering, PIXNET

To overcome these challenges, PIXNET decided to move to the cloud. The business turned to Google Cloud to access pay-as-you-go infrastructure and services; use powerful data warehousing, analysis, reporting, and visualization; accelerate the time to build production environments; and eliminate resource issues created by lack of scalability.

“The scalability and flexibility of Google Cloud supported our business aims, while Google’s decision to open a region in Taiwan also presented opportunities to deliver faster response times to our customers,” explains Tseng.

PIXNET engaged a partner to complete the migration to Google Cloud. The business is running all new projects in the cloud and, in the longer term, plans to move its existing services across.

PIXNET is now using [Google Kubernetes Engine](https://cloud.google.com/kubernetes-engine/) (https://cloud.google.com/kubernetes-engine/) to manage and orchestrate containerization clusters for PIXgoods and its content-labeling API; [Compute Engine](https://cloud.google.com/compute/) (https://cloud.google.com/compute/) to deliver virtual machines in a scalable environment; [Cloud Storage](https://cloud.google.com/storage/) (https://cloud.google.com/storage/) to store and archive JavaScript software development kits and advertising creative; an [App Engine](https://cloud.google.com/appengine/) (https://cloud.google.com/appengine/) web framework and cloud computing platform; and a [Cloud SQL](https://cloud.google.com/sql/) (https://cloud.google.com/sql/) relational database.

PIXNET runs a private space in Google Cloud through [Virtual Private Cloud](https://cloud.google.com/vpc/) (https://cloud.google.com/vpc/), with the ability to

shield its backend environment from internet requests. Cloud Pub/Sub

(<https://cloud.google.com/pubsub/>) provides low-latency messaging to enable PIXNET to seamlessly integrate its systems, while Cloud Composer

(<https://cloud.google.com/composer/>) allows the business to author, schedule, and monitor data pipelines that extend to PIXNET's own data center in Taiwan.

BigQuery unlocks data value

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*—Kent Shih, Manager,
Algorithm Engineering,
R&D Center, PIXNET*

The business is warehousing its datasets for
analysis in BigQuery

(<https://cloud.google.com/bigquery/>), while Looker

Studio

(<https://marketingplatform.google.com/about/data-studio/>)

provides the data visualization needed to support
decision making. “BigQuery in particular allows us to
unlock the value of data in delivering highly relevant,
personalized services to our community and in
enabling advertisers to reach consumers with
targeted messages,” says Tseng.

“With Google Cloud, we can handle nearly 7.7 million unique visitors per day, with each business unit able to quickly and easily view performance against key internal indicators,” says Kent Shih, Manager, Algorithm Engineering, R&D Center at PIXNET.

“Meanwhile, Compute Engine runs the algorithms that allow us to test learning resource scheduling models, and we have used these to create an award-winning PIXNET restaurant recommendation system.”

PIXNET is also using Cloud Vision

(<https://cloud.google.com/vision/>) to analyze and classify customer images. In addition, the business is using graphics processing units (GPUs) through Google Cloud Platform (<https://cloud.google.com/>) to support its machine learning capabilities.

Two key functions of ML

“Our intelligent algorithms and our Google Cloud services provide two key functions for us,” explains Shih. “We can predict the gender, ages and interests of our visitors and classify the 18 billion-plus images on our service.” For its restaurant recommendation system alone, the business has created an algorithm that identifies images of individual restaurants and filters the content to exclude irrelevant elements (such as the taker of a selfie at a restaurant).

PIXNET’s algorithms have been trained to recognize terms in articles prepared by bloggers that relate to products or services. The organization can then link these references to the relevant items for sale in PIXgoods.

The business is also using Google Cloud services to capture data about customer behaviors and the devices they access the PIXNET platform from.

“This enables us to broaden the scope and capabilities of our advertising, website marketing, personalization and recommendation systems,” says Shih. “With Google Cloud Platform, we are ideally positioned to extend personalization and mapping to every one of our users.”

Average response times cut

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With Google Cloud in Taiwan, PIXNET has reduced the average response time for customers from 50 milliseconds to 10 milliseconds and accelerated its deployment cycles. The business has cut the time needed to build production environments from three days to one hour.

The business also uses Google Workspace (<https://workspace.google.com/>) cloud-based collaboration and productivity applications, with Gmail (<https://workspace.google.com/products/gmail/>), Calendar (<https://workspace.google.com/products/calendar/>), and Drive (<https://workspace.google.com/products/drive/>) used extensively for communication, scheduling, and document storage and access.

“Google Cloud Platform artificial intelligence, machine learning, and containerization is now the cornerstone of the architecture running our applications and services,” says Tseng. “We see Google continuing to play a key role in the growth and expansion of our business.”

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Reach out to
our team to
see how
Google Cloud
can help your

business.